

## Calculus II 判断级数的收敛性和幂级数的收敛域

考点一: 判断常数项级数  $\sum_{n=1}^{\infty} (-1)^n a_n$ ,  $a_n > 0$  的收敛性. Section 10.2-10.6

(i) 注意: 收敛包括绝对收敛和条件收敛.

有时候尽管  $a_n > 0$ ,  $a_n \rightarrow 0$ ,  $a_n$  单调递减, 但不是条件收敛, 如  $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$ .

而条件收敛需要满足两个得分点: (1)  $a_n > 0$ ,  $a_n \rightarrow 0$ ,  $a_n$  单调递减; (2) 不绝对收敛.

(ii) 比式判别法必须  $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} > 1$  严格大于1, 若极限等于1, 大部分是  $\lim_{n \rightarrow \infty} a_n \neq 0$ , 从而级数发散. 如:  $\sum_{n=1}^{\infty} (-1)^n (1 - \frac{1}{n})^n$ .

1. (2018年期末) Which of the following series converge absolutely, which converge conditionally, and which diverge? Give reasons for your answer.

$$(1) \sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n(n+1)}} \quad (2) \sum_{n=2}^{\infty} (-1)^n \frac{1}{n(\ln n)^3} \quad (3) \sum_{n=1}^{\infty} (-1)^n \frac{n^2+1}{2n^2+n-1} \quad (4) \sum_{n=1}^{\infty} \frac{(-3)^n}{n!}$$

2. (2019年期中) Does the following series absolutely converge, conditionally converge, or diverge? Give reasons for your answer.

$$(1) \sum_{n=1}^{\infty} (-1)^n (1 - \frac{1}{n})^{n^2} \quad (2) \sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}} \quad (3) \sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2-2019n+1}}$$

3.(2019年期末) The sequences  $\{a_n\}$  and  $\{b_n\}$  satisfy  $0 < a_n < \frac{\pi}{2}$ ,  $0 < b_n < \frac{\pi}{2}$ , and  $\cos a_n - a_n = \cos b_n$ . The series  $\sum_{n=1}^{\infty} b_n$  converges. Show that  $\lim_{n \rightarrow 0} a_n = 0$ .

4.(2021年期中) Determine if the series,  $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p(\ln n)^2}$  ( $p > 0$ ), converges absolutely, or converges conditionally, or diverges. Give reasons for your answer.

5. (2023年期中) Assume that  $a$  and  $b$  are real numbers, and  $0 < b < a < 1$ .

(a) Determine whether the series  $\frac{a}{\sqrt{(1+a)(1+b)}} + \frac{b^2}{\sqrt{(1+b^2)(1+b^2)}} + \frac{a^3}{\sqrt{(1+a^3)(1+b^3)}} + \frac{b^4}{\sqrt{(1+a^4)(1+b^4)}} + \dots + \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} + \dots$ , where  $c_n = a^n$ , if  $n$  is odd;  $c_n = b^n$ , if  $n$  is even.

converges or diverges. Justify your answer!

(b) Find the limit:  $\lim_{n \rightarrow \infty} (\frac{a^n}{n^2} + \frac{b^n}{n})^{1/n}$ .

考点二: 求幂级数的收敛域, 收敛半径.

注意: 一定要考虑端点的收敛性.

1. (2023年期末) (1) Find the interval of convergence of the series  $\sum_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{(n+2023) \ln n}$   
 (2) For what values of  $x$  does the series converge absolutely, or conditionally?
2. (2023年期中) (1) Find the interval of convergence of the series  $\sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n$ .  
 (2) For what values of  $x$  does the series converge absolutely, or conditionally?
3. (2021年期中) For the power series  $f(x) = \sum_{n=1}^{\infty} \frac{n+2}{n(n+1)} x^n$ ,  
 (1) For what values of  $x$  does the power series converge?  
 (2) Find the sum of the series within the interval of convergence.
4. (2020年期末) Find the interval of convergence of the power series  $\sum_{n=0}^{\infty} \frac{2^n}{\ln(n+2)} x^n$ .
5. (2019年期末) (1) Find the interval of convergence of the series  $\sum_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{\sqrt{n+9012} \ln n}$ .  
 (2) For what values of  $x$  does the series converge absolutely, or conditionally?
6. (2019年期中) (1) Find the radius and interval of convergence of the series  $\sum_{n=0}^{\infty} \frac{(-1)^n 2^n x^n}{\sqrt{n^2+n+1}}$ .  
 (2) For what values of  $x$  does the series converge absolutely, or conditionally?